Transparent Medical Image Classification via Explainable AI
Results, Discussion & Conclusion

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1. Quantitative Results

The trained ResNet-50 model achieved strong classification performance on the held-out test set of 624 images. The key metrics are as follows:

- AUC (Area Under the ROC Curve): 0.9701
- PR-AUC (Area Under the Precision-Recall Curve): 0.9796
- Optimal classification threshold: 0.5 (selected based on F1 score maximisation during threshold analysis)

The high PR-AUC of 0.9796 is particularly significant given the class imbalance in the dataset — approximately 73% pneumonia cases in training. PR-AUC is a more informative metric than AUC in imbalanced settings as it directly measures the trade-off between precision and recall for the minority class.

At the default threshold of 0.5, the model correctly identified the majority of pneumonia cases while maintaining low false positive rates. However, threshold analysis revealed 3 false negatives out of 234 pneumonia test cases (recall = 0.987) — pneumonia cases the model failed to detect — which became the primary focus of the explainability analysis in subsequent sections.

The model demonstrates strong performance, approaching levels reported in prior studies, but further validation on external datasets is required before clinical applicability can be claimed. Classification accuracy alone is insufficient for clinical deployment — the following sections evaluate whether the model's reasoning process can be trusted.

2. Qualitative Results — XAI Findings

2.1 Grad-CAM

Grad-CAM analysis revealed that the model's spatial attention varied significantly across cases. For correctly classified pneumonia cases, Grad-CAM highlighted the lower and central lung fields — consistent with the anatomical regions affected by bacterial consolidation. For normal cases, activation was diffuse and low, indicating no dominant focus region.

However, two important limitations were observed. First, on high-confidence pneumonia predictions (p = 1.000), Grad-CAM produced near-zero variance activation maps — a phenomenon known as gradient saturation. This means Grad-CAM failed to produce a meaningful explanation precisely when the model was most certain. Second, in misclassified normal cases, Grad-CAM highlighted the lower lung and abdominal region — not a clinically relevant area for pneumonia diagnosis — suggesting spatial bias in the model's attention mechanism.

(Figure 1 — Grad-CAM heatmap overlay)

Figure 1 shows activation concentrated in the lower lung region for pneumonia cases, with near-zero activation for normal cases.

2.2 SHAP

SHAP analysis provided directional attribution maps showing which regions pushed the prediction toward or away from pneumonia. For correctly classified pneumonia cases, SHAP identified strong positive contributions in the central and lower lung fields — consistent with consolidation patterns in bacterial pneumonia. For normal cases, SHAP assigned predominantly negative values across the image, correctly suppressing the pneumonia prediction.

In false negative cases, SHAP revealed a decision boundary conflict — the model detected some pneumonia signal in the upper lung fields but was simultaneously suppressed by strong negative contributions from other regions, resulting in a net prediction below the 0.5 threshold. This mechanistically explains why these cases were missed.

(Figure 2 — SHAP contribution map)

Figure 2 shows positive red contributions in the central lung fields for pneumonia and predominantly negative blue values for normal cases.

2.3 LIME

LIME identified coarse superpixel regions contributing to each prediction. For pneumonia cases, LIME highlighted the right lung field as the dominant contributing segment. For normal cases, LIME suppressed lung field segments and highlighted peripheral regions. However, LIME exhibited high variance due to its perturbation-based sampling — a known limitation of segment-based explanation methods.

(Figure 3 — LIME segment explanation)

Figure 3 shows highlighted superpixel segments in the right lung field for pneumonia and peripheral suppression for normal cases.

2.4 False Negative Analysis

The three false negative cases were analysed using all three XAI methods simultaneously. A consistent pattern emerged across all methods: the model failed when pneumonia was distributed across multiple lung regions rather than concentrated in a single high-contrast area. This suggests the model has developed a bias toward localised features — a known risk when training on datasets dominated by bacterial pneumonia cases with dense focal consolidations.

3. Comparative Results — Cross-Method Agreement Analysis

The three XAI methods were applied to identical images to evaluate cross-method agreement — the core research contribution of this project.

(Figure 4 — three-way XAI comparison grid)

Figure 4 presents all three methods side by side on identical images, enabling direct visual comparison of spatial agreement and disagreement.

Table 1: Cross-Method XAI Agreement Summary

Case	Prediction	Grad-CAM	SHAP	LIME	Agreement
Normal	P = 0.105	Low/diffuse	Negative	Suppressive	Strong Agreement
Pneumonia	P = 1.000	Flat (saturated)	Positive	Positive	Partial Agreement
FN Case 1	P = 0.043	Peripheral	Negative	Peripheral	Strong Agreement
FN Case 2	P = 0.215	Lower left	Suppressed	Peripheral	Partial Agreement
FN Case 3	P = 0.098	Lateral wall	Negative	Peripheral	Strong Agreement

3.1 Agreement Cases

For the normal image (p = 0.105), all three methods were consistent:
- Grad-CAM showed diffuse low activation across lung fields
- SHAP assigned predominantly negative values across the image

- LIME highlighted peripheral and shoulder regions as suppressive

This three-way agreement provides higher confidence that the model's reasoning is reliable for this case. When all methods point to the same conclusion, the explanation can be considered more faithful to the model's actual decision process.

3.2 Disagreement Cases

For the pneumonia image (p = 1.000), a critical conflict emerged between methods:
- SHAP showed strong positive contributions in the central and lower lung fields — consistent with clinical anatomy
- LIME identified lung field segments as positively contributing
- Grad-CAM produced a completely flat activation map due to gradient saturation

To quantify this disagreement, Pearson correlation was computed between the Grad-CAM and SHAP maps. For the normal image, r = 0.4236, indicating moderate agreement. For the pneumonia image, Grad-CAM produced a near-zero variance map, making correlation mathematically undefined and confirming complete attribution collapse.

(Figure 5 — full XAI pipeline demo)

Figure 5 shows the complete pipeline output — prediction, Grad-CAM, SHAP, and LIME — for a single pneumonia case.

3.3 Trust Evaluation

These findings support the central hypothesis of this project:

"Agreement between XAI methods can serve as a proxy for trust in medical AI systems."

When all three methods agree → higher confidence in explanation reliability.
When methods disagree → the model's decision should be treated with caution.

Critically, disagreement between XAI methods does not always indicate a wrong prediction — the pneumonia case was correctly classified at p = 1.000 despite Grad-CAM's failure. The model was correct. The explanation was not. This mismatch raises serious concerns for clinical deployment of single-method interpretability pipelines.

4. Discussion

4.1 Model Performance & Clinical Relevance

The model achieved AUC = 0.9701 and PR-AUC = 0.9796, demonstrating clinically competitive performance on the chest X-ray pneumonia detection task. However, the presence of 3 false negatives — cases where pneumonia was missed entirely — highlights that high aggregate performance does not guarantee reliability at the individual case level. In clinical settings, a single missed pneumonia diagnosis can have serious consequences, making false negative analysis critical.

4.2 Spatial Bias in Model Attention

Grad-CAM analysis revealed that the model occasionally attends to non-anatomical regions — including the lower lung border and abdominal area — when misclassifying normal cases as pneumonia. This spatial bias suggests the model may have learned dataset-specific artifacts rather than purely pathological features. This is consistent with findings in the literature — Arun et al. (2021) demonstrated that explanation maps frequently fail to align with radiologist annotations, even when classification accuracy is high.

4.3 Decision Boundary Conflict

SHAP analysis of false negative cases revealed a decision boundary conflict — the model detected pneumonia signal in some lung regions but was simultaneously suppressed by negative contributions from other regions. This conflict, confirmed independently by LIME, provides a mechanistic explanation for why these cases were missed. It suggests the model is sensitive to the spatial distribution of disease — performing well on focal consolidations but struggling with diffuse or bilateral patterns.

4.4 XAI Method Reliability

The comparative analysis demonstrated that different XAI methods capture fundamentally different aspects of model behaviour:

- Grad-CAM systematically fails to capture diffuse pathological patterns and suffers from gradient saturation on high-confidence predictions. It should not be used as the sole explanation method in clinical deployment.

- SHAP demonstrated the most consistent behaviour in this study, suggesting it may be more suitable for clinical audit scenarios compared to gradient-based methods.

- LIME provides intuitive segment-level explanations but exhibits high variance due to perturbation-based sampling. It is useful for demonstration but less reliable for rigorous evaluation.

4.5 Explanation Faithfulness & Trust

A critical finding of this project is that model correctness does not guarantee explanation reliability. The pneumonia case was correctly classified at p = 1.000, yet Grad-CAM produced a completely flat activation map — providing no useful explanation despite the correct prediction. This highlights the issue of explanation faithfulness and raises concerns about relying on a single XAI method in safety-critical applications.

To our knowledge, this is one of the first works to explicitly evaluate cross-method agreement between Grad-CAM, SHAP, and LIME as a proxy for trust in medical imaging models.

These findings reinforce the importance of evaluating explanation consistency across multiple methods. This project proposes cross-method agreement as a validation step before clinical trust is established. This framework has not been formally explored in prior chest X-ray classification studies.

4.6 Limitations

Several limitations of this work should be acknowledged:

- The XAI comparison was conducted on a limited number of test images. Statistical generalisation requires evaluation across a larger dataset.
- LIME's instability means results may vary across runs, limiting reproducibility.
- The dataset is dominated by bacterial pneumonia cases, which may bias the model toward localised consolidation patterns and reduce performance on viral or atypical pneumonia presentations.
- The trust evaluation framework proposed here is qualitative. Future work should formalise this framework by defining quantitative agreement metrics — such as correlation scores and spatial overlap measures — and thresholds for trust calibration.

5. Conclusion

This project presented a multi-method explainability pipeline for chest X-ray pneumonia classification using ResNet-50. The pipeline combines Grad-CAM, SHAP, and LIME to provide complementary explanations of model decisions, and introduces cross-method agreement as a framework for evaluating trust in medical AI systems.

The model achieved strong classification performance with AUC = 0.9701 and PR-AUC = 0.9796. However, the primary contribution of this work is not the classification performance — it is the systematic analysis of when and why the model's explanations can be trusted.

Three key findings emerged from this analysis:

First, Grad-CAM systematically fails to capture diffuse pathological patterns and suffers from gradient saturation on high-confidence predictions — making it unreliable as a standalone explanation method in clinical settings.

Second, SHAP demonstrated the most consistent behaviour in this study, suggesting it may be more suitable for clinical audit scenarios compared to gradient-based methods.

Third, and most importantly — disagreement between XAI methods is a strong indicator of unreliable model reasoning. A model can be correct for the wrong reasons, and only a multi-method XAI approach can expose this.

This finding directly supports the central hypothesis of this project:

"Agreement between XAI methods can serve as a proxy for trust in medical AI systems."

This work demonstrates that explainability in medical AI should not be treated as a single-visualisation problem, but as a multi-perspective validation process. Future work should evaluate cross-method agreement quantitatively across larger datasets, incorporate radiologist feedback to validate explanation faithfulness, and extend the framework to other medical imaging modalities and architectures.
In high-stakes medical AI, agreement between explanations may matter more than accuracy alone. A model that is correct without reliable reasoning is not clinically trustworthy.

The code, notebooks, and results are publicly available at:

https://github.com/emanaak04-svg/medical-xai